BTEUP EXAM NOTES | HINGLISH EDITION

Information Security

Unit 1: Introduction, PAIN, OS Security, RAID and Anti-Virus

In notes ke baare me

Part A me har topic simple Hinglish me real-life example ke saath samjhaya hai, taaki beginner bhi zero se samajh sake.

Part B me important exam questions ke poore 5-marks wale answers Hinglish me diye hain.

Last me: quick revision sheet, common mistakes aur self-test.

Exam me answer English me likhna hai to English Edition PDF use karo.

Syllabus covered (Unit 1)

Introduction to Information Security, Various aspects of information security (PAIN), Security Features of Operating Systems: Authentication, Logs, Audit Features, File System Protection, User Privileges, RAID options, Anti-Virus Software.

Unit Overview aur Contents

Simple idea: Information Security ka matlab hai data ko galat logon se bachana, aur ye pakka karna ki sahi log use sahi time par use kar saken. Is unit me 3 cheezein hain: (1) basics aur PAIN, (2) Operating System ke security features, (3) RAID aur Anti-virus se extra protection.

Topic	Exam importance	Notes me kahan
Information Security, CIA Triad	High	Part A: Topic 1 | Part B: Q1, Q9
PAIN (4 aspects)	Very High (lagbhag pakka)	Part A: Topic 2 | Part B: Q1, Q12
OS Security: Authentication, Logs, File protection, Privileges	Very High	Part A: Topic 3 | Part B: Q3 se Q8
RAID options	Very High	Part A: Topic 4 | Part B: Q2, Q10
Anti-Virus Software	Very High	Part A: Topic 5 | Part B: Q4, Q11

Contents

Part A: Easy Explanation	Part B: Important Questions aur Answers
Topic 1: Information Security aur CIA Triad Topic 2: PAIN Topic 3: OS Security Features Topic 4: RAID Topic 5: Anti-Virus Software	Q1 Information Security aur PAIN (Very Likely) Q2 RAID aur uske levels (Very Likely) Q3 OS security features (Very Likely) Q4 Anti-virus software (Very Likely) Q5 Authentication (Likely) Q6 Logs aur Audit (Likely) Q7 File system protection (Likely) Q8 User privileges (Likely) Q9 CIA Triad (Likely) Q10 RAID 0 vs RAID 1 (Possible) Q11 Signature vs Heuristic (Possible) Q12 Non-repudiation (Possible)
Uske baad: Quick Revision Sheet, Common Mistakes, Self-Test answers ke saath	

Is unit ko kaise padhein (time: 1 din)

1) Part A ek baar padh kar samjho. 2) Part B ke answers diagram ke saath copy me likh kar practice karo. 3) Quick Revision Sheet se revise karo. 4) Self-Test bina dekhe karo.

PART A: Easy Explanation (Hinglish)

Topic 1: Information Security and CIA Triad

Real-life example

Socho tumhare ghar ki almirah me paise rakhe hain. Almirah par **tala** lagana security hai. Information Security bhi data ko aise hi tala lagati hai, taaki koi galat insaan use dekh, badal ya delete na kar sake.

Definition

Information Security (InfoSec) is the practice of protecting information from unauthorized access, use, disclosure, modification or destruction.

Key points

- Ye sirf computer data ki nahi, **har tarah ki information** (paper, digital, bolkar) ki security hai.
- Iske **3 main goals** hain, jinhe **CIA Triad** kehte hain.
- **Confidentiality:** data sirf authorized log dekh saken. Example: bank password sirf tumhe pata ho.
- **Integrity:** data bina permission ke change na ho. Example: marksheet me marks koi na badle.
- **Availability:** data jab chahiye tab mile. Example: bank ka server 24x7 chalu rahe.

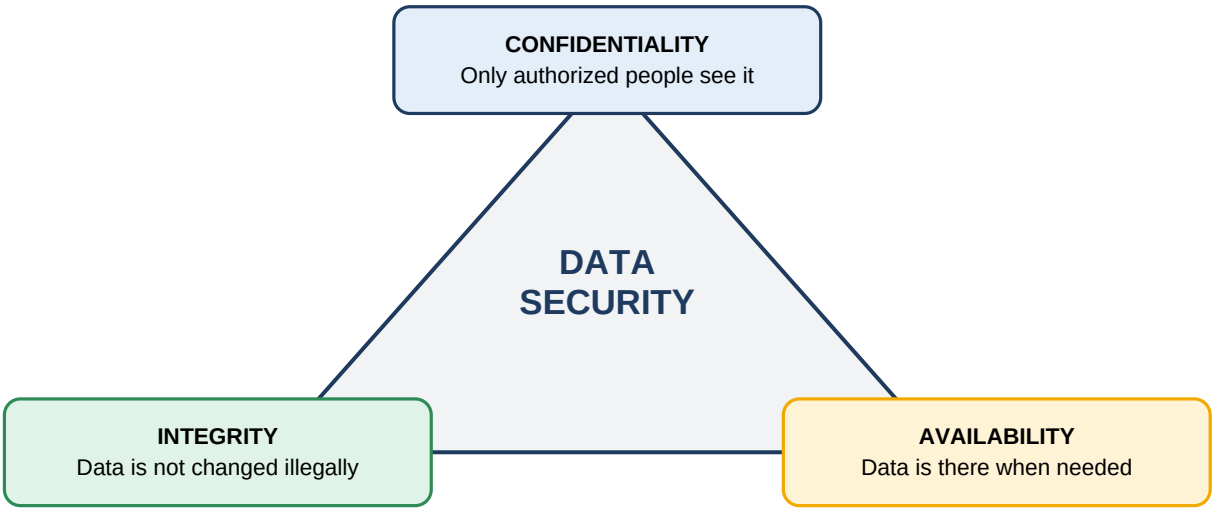


Figure 1: CIA Triad

Topic 2: PAIN (Aspects of Information Security)

Real-life example

Maan lo tum online payment karte ho. System ko 4 cheezein pakki karni hain: tumhara data koi dekh na sake (**Privacy**), payment tum hi kar rahe ho (**Authentication**), amount raste me badla na ho (**Integrity**), aur baad me tum "maine nahi kiya" na bol sako (**Non-repudiation**).

Definition

PAIN stands for **P**rivacy, **A**uthentication, **I**ntegrity and **N**on-repudiation. These are the four main aspects (pillars) of information security.

Aspect	Matlab	Kaise milta hai	Example
Privacy	Data sirf authorized person dekh sake	Encryption, access control	WhatsApp end-to-end encryption
Authentication	User ki pehchan verify karna	Password, OTP, biometrics	ATM PIN
Integrity	Data transit ya storage me change na ho	Hash, checksum, digital signature	File download ke baad hash match karna
Non-repudiation	Bhejne wala baad me mukar na sake	Digital signature	Online payment ke baad inkaar nahi kar sakte

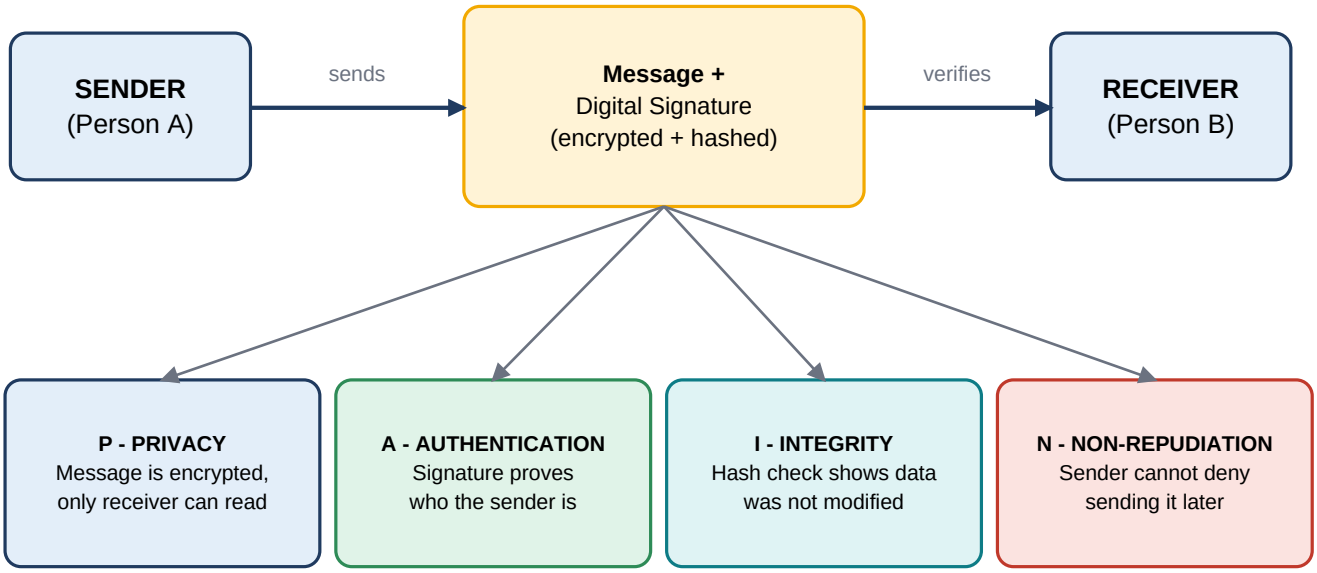


Figure 2: PAIN with Digital Signature flow

Yaad rakho

Privacy ko kabhi kabhi **Confidentiality** bhi kehte hain. Authentication aur **Authorization** alag cheezein hain: Authentication = "tum kaun ho?", Authorization = "tumhe kya karne ki permission hai?"

Topic 3: Security Features of Operating Systems

Operating System (Windows, Linux, Android) computer ka manager hai. Isliye uski security sabse pehli defence line hoti hai. OS ke **4 mukhya security features** hain:

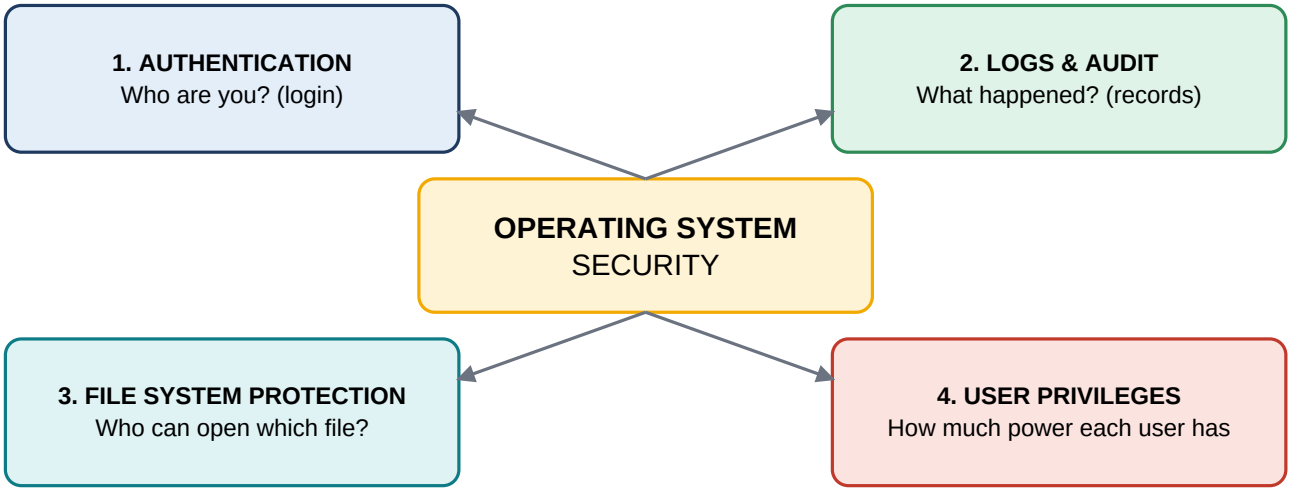


Figure 3: OS Security Features

3(a) Authentication

> **Real-life example**
> Bank me paise nikalne jate ho to staff **ID proof** maangta hai. Computer bhi login ke time pehchan maangta hai.

Definition: Authentication is the process of verifying the identity of a user before giving access to the system.

- **Something you know:** Password, PIN.
- **Something you have:** OTP, smart card, token.
- **Something you are:** Fingerprint, face, iris (biometrics).
- 2 ya zyada factors ek saath use karne ko **Multi-Factor Authentication (MFA)** kehte hain. Example: ATM card + PIN.

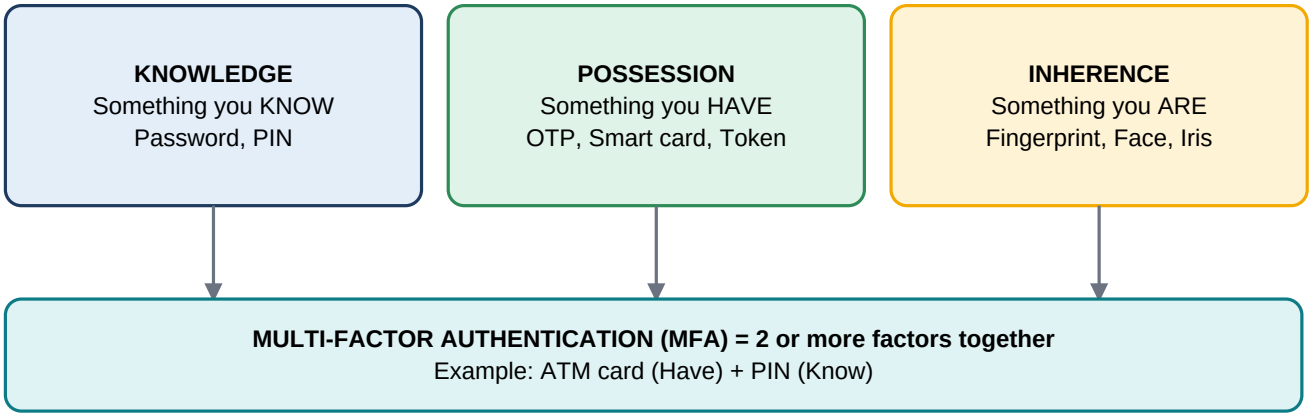


Figure 4: Authentication factors

3(b) Logs and Audit Features

> **Real-life example**
> CCTV camera jaise dukaan ki har activity record karta hai, waise hi **logs** computer ki har activity record karte hain.

Log: A record of events happening in the system (who logged in, when, what was changed). **Audit:** Examining these logs to find suspicious or illegal activity.

- Failed login attempts, file access, program errors aur system changes record hote hain.
- Hacking ke baad logs se pata chalta hai **kya hua, kab hua, kisne kiya**.
- **Windows:** Event Viewer. **Linux:** `/var/log/` folder.
- Types: **Security log, System log, Application log**.

3(c) File System Protection

> **Real-life example**
> Ghar me har kamre ki alag chabi hoti hai. Har file/folder ke liye bhi alag permission hoti hai.

Definition: File system protection controls who can read, write or execute a file or folder.

- **Permissions:** Read (r), Write (w), Execute (x).
- **Linux:** permissions Owner, Group, Others ke liye alag hoti hain. Command: `chmod`.
- **Windows (NTFS): ACL (Access Control List)** decide karti hai kaun kya kar sakta hai.
- **Encryption:** Windows me **EFS** (file level) aur **BitLocker** (poori disk) data ko lock karte hain.

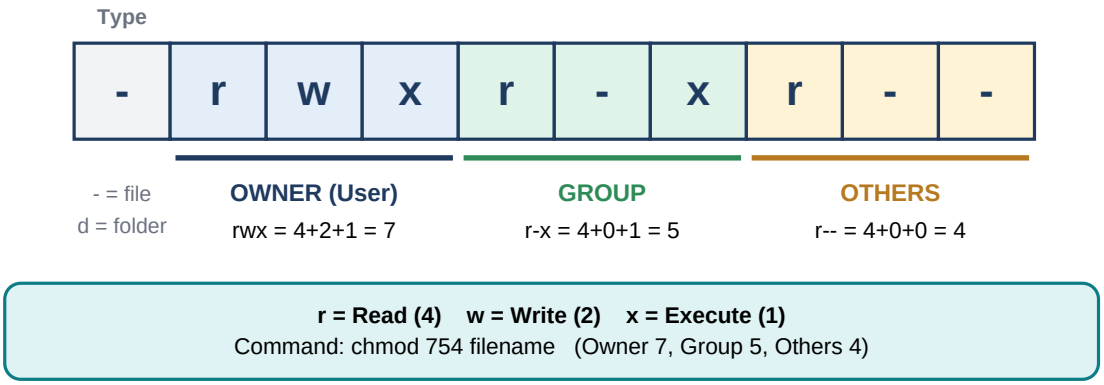


Figure 5: Linux file permission -rwxr-xr--

3(d) User Privileges

> **Real-life example**
> Office me manager ke paas har cabinet ki chabi hoti hai, par peon ke paas sirf kuch cabinets ki. Isse galti ya chori ka nuksan kam hota hai.

Definition: User privileges are the rights and permissions given to a user account to perform actions on the system.

- **Principle of Least Privilege:** user ko utne hi rights do jitne kaam ke liye zaroori hain.
- **Administrator / root:** full control. **Standard user:** limited control. **Guest:** bahut kam access.
- **Windows UAC** (User Account Control): system change se pehle permission maangta hai.
- **Linux sudo:** temporary admin power deta hai.
- Benefit: malware ya galti se hone wala damage kam hota hai.

Topic 4: RAID Options

Real-life example

Maan lo tumhare paas ek hi notebook hai aur wo kho jaye to sab notes gaye. Agar tum **2 notebooks me same notes** likho to ek kho jaye tab bhi notes bach jayenge. RAID isi tarah **kai hard disks** ko mila kar data safe ya fast banata hai.

Definition

RAID (Redundant Array of Independent Disks) is a technique that combines multiple physical disks into one logical unit to improve performance and/or data protection.

3 basic techniques

- **Striping:** data ko disks me tukdon me baantna. Speed badhti hai.
- **Mirroring:** same data ki copy dusri disk par. Safety milti hai.
- **Parity:** extra calculated data, jisse kharab disk ka data wapas ban sakta hai.

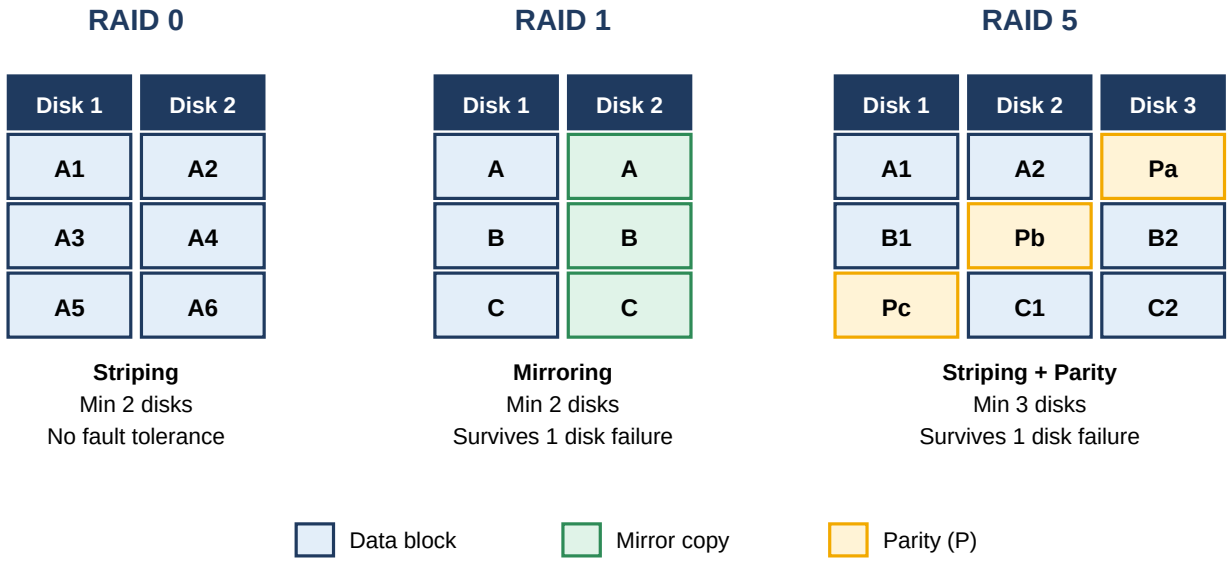


Figure 6: RAID 0, RAID 1 and RAID 5 layouts

RAID Level	Technique	Min disks	Fault tolerance	Fayda / Nuksan
RAID 0	Striping	2	None (1 disk fail = all data lost)	Very fast, but not safe
RAID 1	Mirroring	2	1 disk can fail	Safe, but 50% space wasted
RAID 5	Striping + distributed parity	3	1 disk can fail	Good balance of speed and safety
RAID 6	Striping + double parity	4	2 disks can fail	Safer, slightly slower
RAID 10	Mirroring + striping	4	1 disk per mirror pair	Fast and safe, but expensive

Bahut important

RAID backup ka replacement nahi hai. Agar file delete ho gayi ya virus aa gaya, RAID usse nahi bachata, kyunki galat data bhi sabhi disks par copy ho jata hai. Backup alag se rakhna zaroori hai.

Topic 5: Anti-Virus Software

Real-life example

Anti-virus computer ka **chowkidar** hai. Jaise guard gate par har aane wale ko check karta hai, waise anti-virus har file ko check karta hai.

Definition

Anti-virus software is a program that detects, blocks and removes malware such as viruses, worms and trojans.

Detection methods (kaise kaam karta hai)

- **Signature-based:** known viruses ki pehchan (signature) database se match karta hai. Naye virus pakad nahi pata.
- **Heuristic-based:** program ke suspicious code ya style ko dekh kar anumaan lagata hai.
- **Behaviour-based:** program chalte waqt kya karta hai (jaise files encrypt karna) wo monitor karta hai.
- **Sandboxing:** suspicious file ko alag safe jagah me chala kar test karta hai.

Main features

- **Real-time protection:** background me hamesha scan.
- **Full / scheduled scan:** poore system ki jaanch.
- **Quarantine:** infected file ko alag, safe folder me band karna.
- **Auto-update:** virus definitions regularly update hoti hain.
- Email aur web protection. Examples: Windows Defender, Quick Heal, Kaspersky, Norton.

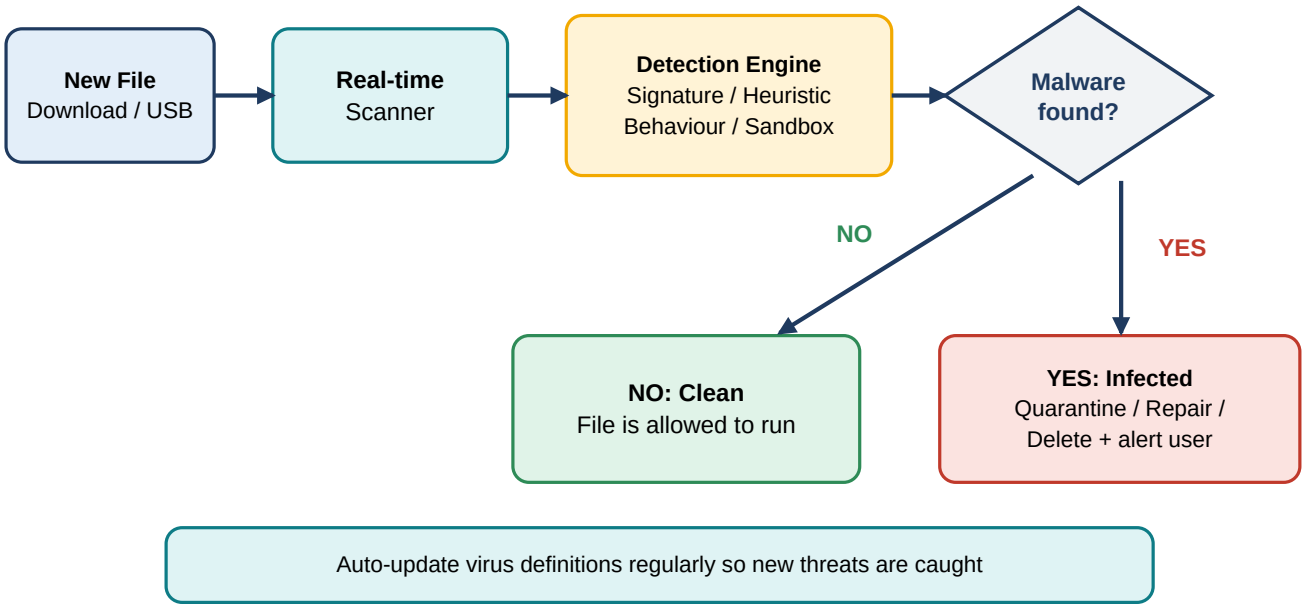


Figure 7: Working of Anti-virus software

Limitation

Zero-day (bilkul naye) viruses pehle pakad me nahi aate, isliye regular update zaroori hai.

PART B: Important Questions aur Model Answers (Hinglish)

Priority: **Very Likely** = BTEUP me sabse zyada aata hai. **Likely** = achha chance. **Possible** = aa sakta hai. Har answer exam format me hai: **Definition, Points, Diagram/Table, Conclusion**.

Q1. Information Security kya hai? PAIN aspects explain karo. — VERY LIKELY

Definition / Parichay: Information Security ka matlab hai information ko unauthorized access, use, disclosure, modification ya destruction se bachana. Iske main aspects ko **PAIN** kehte hain: Privacy, Authentication, Integrity aur Non-repudiation.

Points

- **Privacy:** information sirf authorized logon ko milti hai. Ye encryption aur access control se hota hai.
- **Authentication:** verify karna ki user ya sender sach me wahi hai jo wo bol raha hai. Password, OTP ya biometrics se.
- **Integrity:** information storage ya transmission me badli na jaye. Hash function, checksum aur digital signature se.
- **Non-repudiation:** sender baad me ye inkaar nahi kar sakta ki usne message ya transaction kiya tha. Digital signature se milta hai.

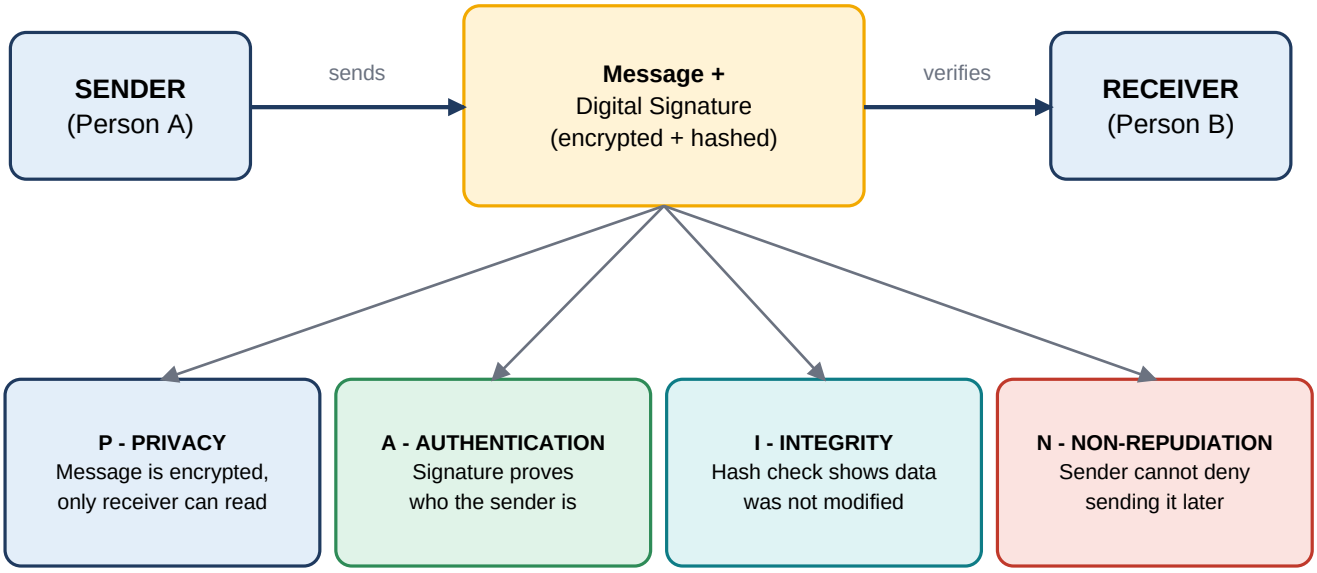


Figure: PAIN with digital signature

Conclusion: PAIN milkar ensure karta hai ki information secret, genuine, unchanged aur legally accountable rahe.

Q2. RAID kya hai? RAID level 0, 1 aur 5 explain karo. VERY LIKELY

Definition / Parichay: RAID (Redundant Array of Independent Disks) ek technology hai jisme kai physical disks ko ek logical unit me jodte hain, taaki performance, fault tolerance ya dono badh sakein.

Points

- **Striping** data ko disks me baant kar speed badhata hai, **mirroring** same copy rakhta hai, aur **parity** extra data rakhta hai jisse kharab disk wapas banayi ja sake.
- **RAID 0 (Striping):** minimum 2 disks. Bahut fast, par fault tolerance nahi. Ek disk fail hui to sara data gaya.
- **RAID 1 (Mirroring):** minimum 2 disks. Dono disks par same data. Ek disk fail ho sakti hai, par sirf 50% capacity use hoti hai.
- **RAID 5 (Striping + distributed parity):** minimum 3 disks. Ek disk fail ho sakti hai. Speed, safety aur space ka achha balance.
- RAID 6 me double parity hoti hai (min 4 disks, 2 failures bardasht). RAID 10 me mirroring aur striping dono hote hain (min 4 disks).

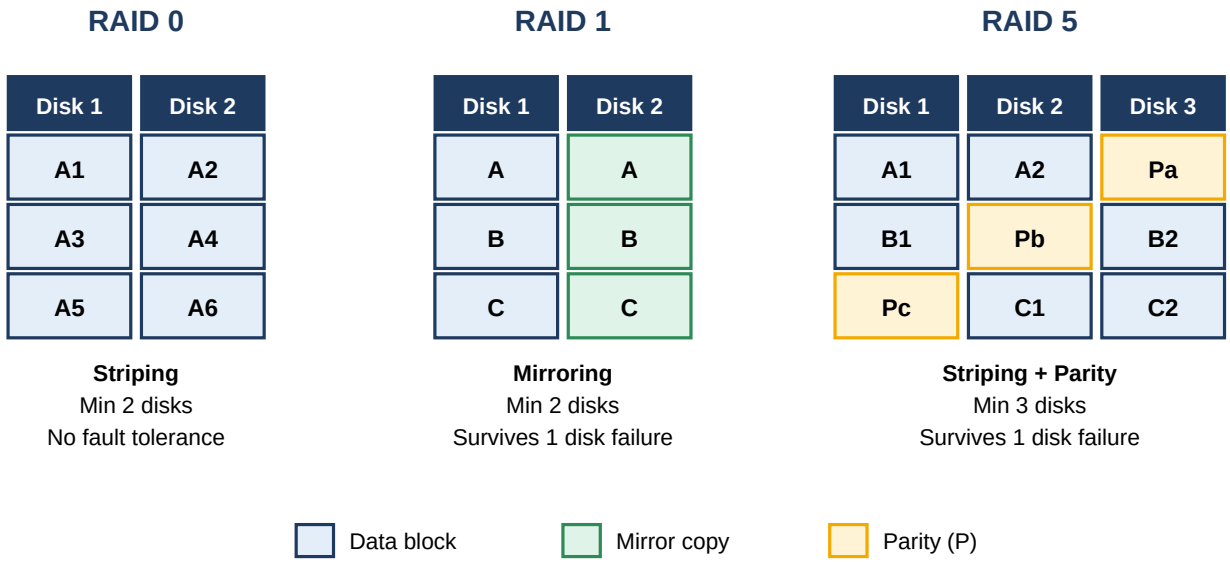


Figure: RAID 0, 1 aur 5

Level	Min disks	Fault tolerance	Speed	Space use
RAID 0	2	Koi nahi	Bahut high	100%
RAID 1	2	1 disk	Read achhi	50%
RAID 5	3	1 disk	High	(n-1)/n

Conclusion: RAID data ki speed aur availability badhata hai, par regular backup ka replacement nahi hai.

Q3. Operating System ke security features explain karo. VERY LIKELY

Definition / Parichay: Operating System security ka matlab hai system resources (files, memory, devices) ko unauthorized access aur misuse se bachana. Modern OS ye features deta hai.

Points

- **Authentication:** login se pehle password, PIN, smart card ya biometrics se user ki identity verify karta hai.
- **Logs aur Audit:** logins, file access, errors jaise events record karte hain, taaki suspicious activity trace ho sake.
- **File System Protection:** permissions (read, write, execute), ACL aur encryption (EFS, BitLocker) se files ka access control hota hai.
- **User Privileges:** har user ko sirf kaam ke liye zaroori rights milte hain (least privilege). Example: Windows UAC, Linux sudo.

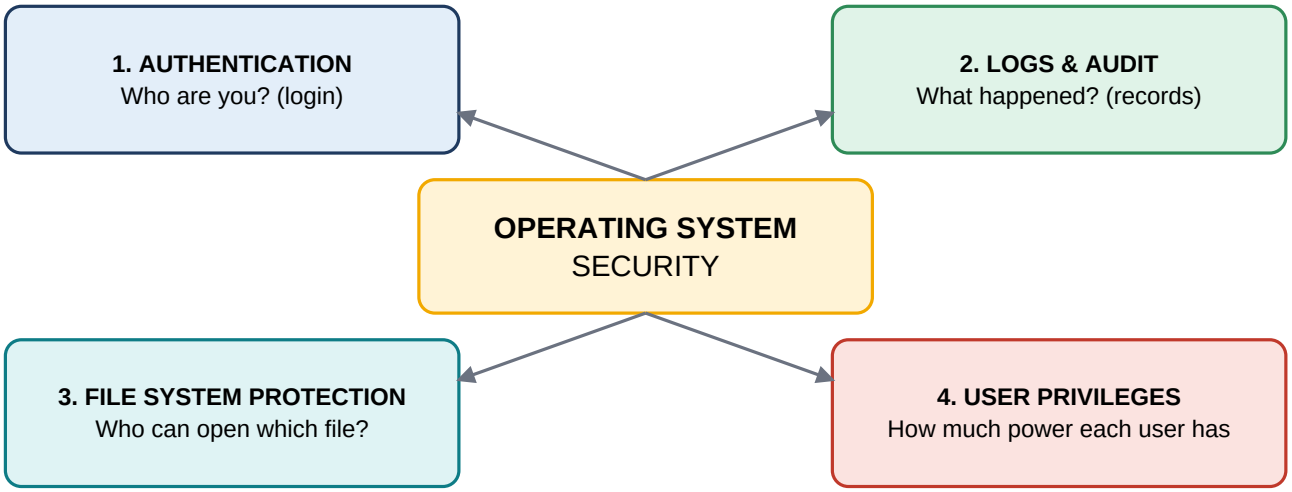


Figure: OS ke security features

Conclusion: Ye features milkar OS me data ki confidentiality, integrity aur availability ki raksha karte hain.

Q4. Anti-virus software kya hai? Ye kaise kaam karta hai? VERY LIKELY

Definition / Parichay: Anti-virus software ek program hai jo virus, worm, trojan aur ransomware jaise malicious software ko detect, prevent aur remove karta hai.

Points

- **Signature-based detection:** files ko known virus signatures ke database se match karta hai.
- **Heuristic detection:** code me suspicious patterns dekh kar unknown viruses pakadta hai.
- **Behaviour-based detection:** chalte hue programs ko monitor karke harmful actions block karta hai.
- **Sandboxing:** suspicious file ko alag isolated environment me chala kar safely observe karta hai.
- **Features:** real-time protection, scheduled scan, quarantine, auto update, email aur web protection.
- **Limitation:** naye (zero-day) threats definitions update hone tak bach sakte hain.

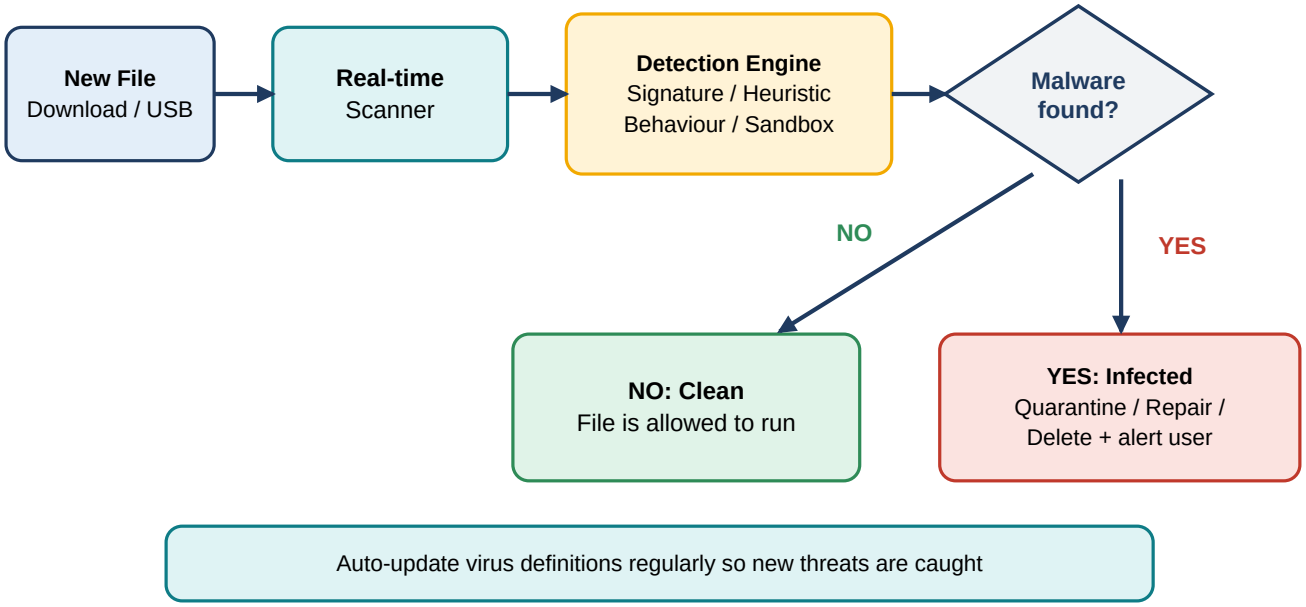


Figure: Anti-virus ka working

Conclusion: Regular update ke saath anti-virus kisi bhi computer system ki ek zaroori protection layer hai.

Q5. Authentication kya hai? Uske types explain karo. LIKELY

Definition / Parichay: Authentication wo process hai jisme system access dene se pehle user, device ya process ki identity verify ki jaati hai.

Points

- **Knowledge factor (kuch jo aap jaante ho):** password, PIN, security question.
- **Possession factor (kuch jo aapke paas hai):** OTP, smart card, hardware token, mobile phone.
- **Inherence factor (kuch jo aap ho):** fingerprint, face, iris ya voice recognition.
- 2 ya zyada factors use karne ko **Multi-Factor Authentication (MFA)** kehte hain. Example: ATM card + PIN.

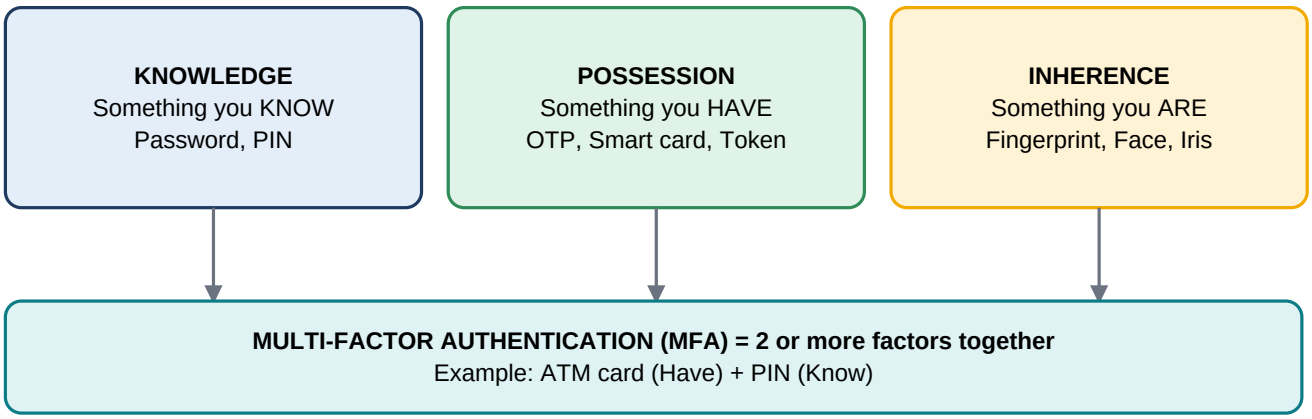


Figure: Authentication factors

Authentication	Authorization
Verify karta hai ki user **kaun** hai	Decide karta hai ki user **kya kar sakta hai**
Pehle hota hai (login)	Authentication ke baad hota hai
Example: password dalna	Example: user file padh sakta hai par delete nahi

Conclusion: Strong authentication, khaas kar MFA, unauthorized access ka risk bahut kam kar deta hai.

Q6. Operating System ke Logs aur Audit features explain karo. LIKELY

Definition / Parichay: Log system me hone wale events ka record hota hai. **Audit** in logs ko check karne ki process hai, taaki security violations aur unusual activity pakdi ja sake.

Points

- Logs me user login, failed login attempts, file access, software installation aur system errors, date, time aur user ke saath record hote hain.
- Audit se attacks detect hote hain, breach ka source milta hai aur accountability prove hoti hai.
- Logs troubleshooting aur legal/compliance requirements me bhi kaam aate hain.
- **Windows** me logs **Event Viewer** me aur **Linux** me `/var/log/` directory me hote hain.
- Logs ko modification se bachana zaroori hai, warna attacker apne nishaan mita sakta hai.

Log type	Kya record karta hai
Security log	Logins, logouts, failed attempts, permission changes
System log	OS events jaise startup, shutdown, driver errors
Application log	Installed programs ke events aur errors

Conclusion: Logging aur auditing se visibility aur accountability milti hai, jo security incidents ko detect aur investigate karne ke liye zaroori hai.

Q7. File System Protection explain karo. LIKELY

Definition / Parichay: File system protection ka matlab hai ye control karna ki kaun se users files aur folders ko read, write ya execute kar sakte hain, taaki unauthorized access ya modification na ho.

Points

- **Permissions:** Linux me read (r), write (w), execute (x) Owner, Group aur Others ko alag alag di jaati hain.
- **ACL (Access Control List):** Windows NTFS me ek list jo batati hai kis user/group ko kis object par kya rights hain.
- **Encryption:** EFS individual files ko aur BitLocker poori drive ko encrypt karta hai.
- **Anya tarike:** files par password, hidden ya read-only attributes, regular backup.

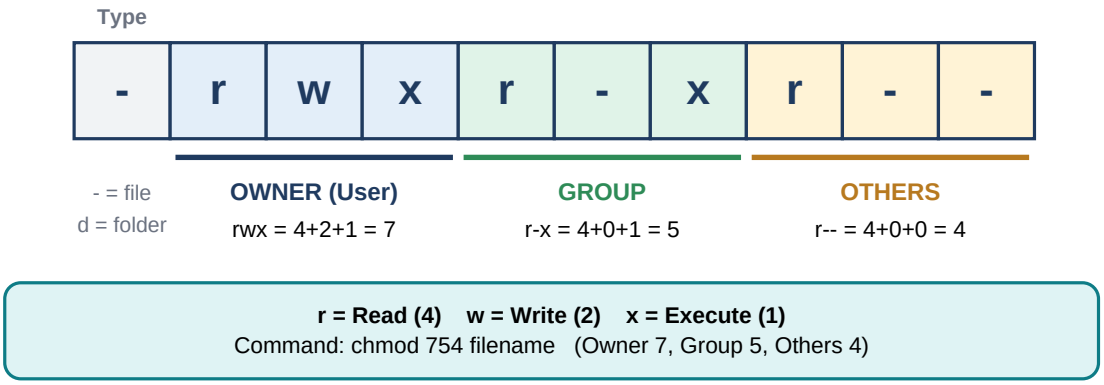


Figure: Linux permission -rwxr-xr-- (chmod 754)

Conclusion: File system protection sirf authorized access dekar data ko confidential aur unmodified rakhta hai.

Q8. User Privileges kya hain? Principle of Least Privilege explain karo. LIKELY

Definition / Parichay: User privileges wo rights aur permissions hain jo user account ko diye jaate hain, aur jo decide karte hain ki user system par kya kar sakta hai.

Points

- **Principle of Least Privilege:** har user ya program ko sirf utna hi access milna chahiye jitna uske kaam ke liye zaroori hai.
- Isse galti, malware aur insider misuse se hone wala nuksan kam hota hai.
- **Windows UAC** administrator rights wale kaam se pehle confirmation maangta hai.
- **Linux sudo** normal users ko temporary administrator rights deta hai.

Account type	Privileges
Administrator / root	Full control: software install, settings change, users manage
Standard user	Programs chalana aur apni files use karna, limited system changes
Guest	Bahut limited, temporary access

Conclusion: Sahi privilege management attack surface kam karta hai aur overall system security badhata hai.

Q9. CIA Triad explain karo. LIKELY

Definition / Parichay: CIA Triad ek model hai jisme information security ke 3 core goals hain: Confidentiality, Integrity aur Availability.

Points

- **Confidentiality:** data sirf authorized users dekh saken. Encryption, passwords, access control se milta hai.
- **Integrity:** data sahi aur unchanged rahe. Hashing, checksums, digital signatures se milta hai.
- **Availability:** data aur systems jab zarurat ho tab mil saken. Backups, RAID, redundancy aur anti-DoS measures se milta hai.

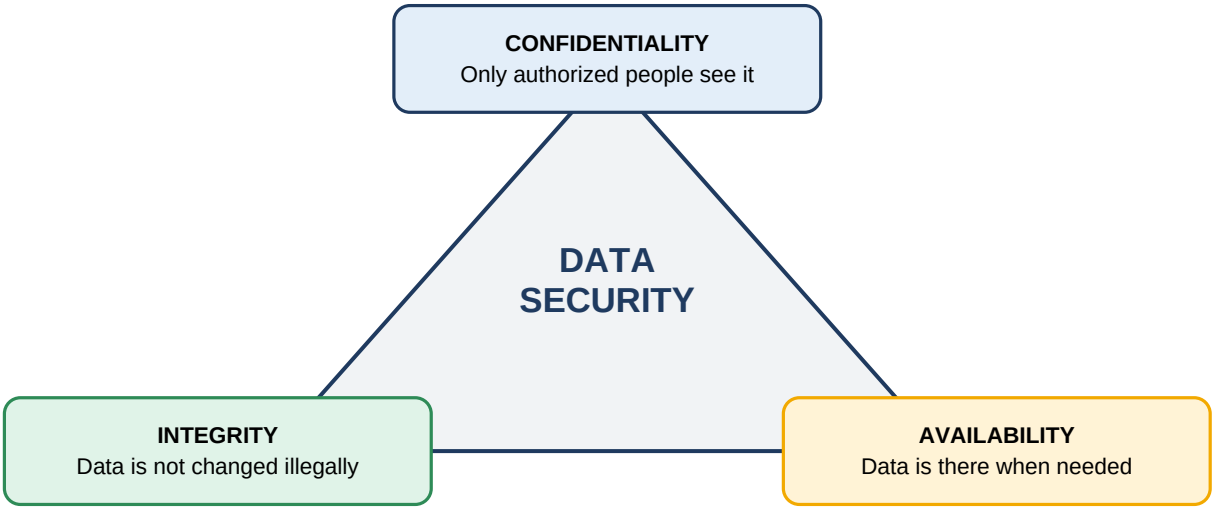


Figure: CIA Triad

Conclusion: Ek secure system ko CIA Triad ke teeno goals ka balance rakhna padta hai.

Q10. RAID 0 aur RAID 1 me difference likho. POSSIBLE

Definition / Parichay: RAID 0 striping se speed badhata hai, jabki RAID 1 mirroring se data ko protect karta hai.

Points

- Dono me minimum do disks chahiye, par unka purpose alag hai.

Point	RAID 0	RAID 1
Technique	Striping	Mirroring
Main goal	High performance	Data safety
Fault tolerance	Koi nahi	Ek disk failure bardasht
Usable capacity	100%	50%
Cost	Kam	Zyada (double disks)
Use case	Video editing, gaming	Important data, servers

Conclusion: Speed chahiye to RAID 0, reliability chahiye to RAID 1 chuna jaata hai.

Q11. Signature-based aur Heuristic-based detection me difference likho. POSSIBLE

Definition / Parichay: Dono anti-virus software ke malware pehchanne ke methods hain.

Points

- Signature-based detection known patterns dhoondhta hai, jabki heuristic detection suspicious behaviour ya code dhoondhta hai.

Point	Signature-based	Heuristic-based
Method	File ko known virus signatures ke database se match karta hai	Code/behaviour ko suspicious traits ke liye analyse karta hai
Detects	Sirf known viruses	Known aur unknown (naye) viruses
Accuracy	Bahut accurate, kam false alarms	False positives ho sakte hain
Update need	Frequent database update chahiye	Updates par kam depend

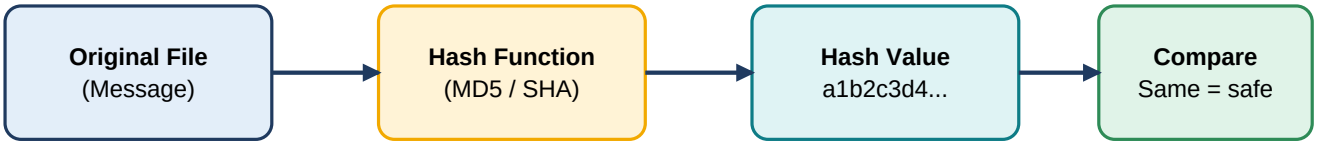
Conclusion: Modern anti-virus behtar protection ke liye dono methods ko mila kar use karta hai.

Q12. Non-repudiation kya hai? Example ke saath samjhao. POSSIBLE

Definition / Parichay: Non-repudiation ensure karta hai ki message bhejne wala ya transaction karne wala baad me usse inkaar na kar sake.

Points

- Ye mainly **digital signatures** se milta hai, jo sender ki private key se bante hain aur public key se verify hote hain.
- Audit logs aur timestamps bhi proof ka kaam karte hain.
- **Example:** agar customer ne OTP aur digital proof ke saath online payment approve kiya, to wo baad me nahi bol sakta ki usne payment kiya hi nahi.
- E-commerce, banking aur legal contracts me ye bahut important hai.



If even 1 character changes, the hash value changes completely (Integrity check)

Figure: Signed message ki integrity prove karne ke liye hash value

Conclusion: Non-repudiation actions ka legal proof deta hai aur digital transactions me trust banata hai.

Quick Revision Sheet (exam se pehle raat ko padho)

Term	Ye yaad rakho
Information Security	Data ko unauthorized access, change, destruction se bachana
CIA	Confidentiality, Integrity, Availability
PAIN	Privacy, Authentication, Integrity, Non-repudiation
Authentication factors	Know (password), Have (OTP, card), Are (fingerprint)
Authentication vs Authorization	Tum kaun ho? vs Tum kya kar sakte ho?
Log / Audit	Log = events ka record. Audit = logs ki jaanch
Windows / Linux logs	Event Viewer / `/var/log/`
Linux permissions	r=4, w=2, x=1. Owner, Group, Others. chmod 754
Windows file security	NTFS ACL, EFS (file), BitLocker (drive)
Least privilege	Sirf minimum zaroori access do
UAC / sudo	Windows admin prompt / Linux temporary admin
RAID full form	Redundant Array of Independent Disks
RAID 0 / 1 / 5	Striping, safety nahi / Mirror, min 2 disks / Parity, min 3 disks
RAID backup nahi hai	Delete hui file ya virus sabhi disks par copy ho jata hai
Anti-virus methods	Signature, Heuristic, Behaviour, Sandbox
Quarantine	Infected file ko safe jagah alag rakhna
Zero-day	Bilkul naya threat jo abhi virus database me nahi hai

Yaad rakhne ke tricks

RAID numbers: 0 = Zero safety, 1 = One copy (mirror), 5 = Parity (minimum **3** disks).
Authentication factors: K-H-A = Know, Have, Are.
PAIN: P-A-I-N. Security ignore karoge to 'pain' hoga.
Anti-virus methods: S-H-B-S = Signature, Heuristic, Behaviour, Sandbox.

Common Mistakes jo Students Karte Hain

- PAIN me **Authentication** ki jagah **Authorization** likh dena. Sahi word Authentication hai.
- Authentication (tum kaun ho?) aur Authorization (tum kya kar sakte ho?) ko mix karna.
- RAID ko backup bolna. Ye galat hai.
- RAID 0 ko safe batana. Isme **koi** redundancy nahi hoti.
- Diagram na banana. 5 marks ke question me neat labelled diagram se marks pakke hote hain.
- Sirf definition likh kar chhod dena. Hamesha points, example aur conclusion bhi likho.
- Examples bhool jana, jaise MFA ke liye ATM card + PIN, anti-virus ke liye Windows Defender.

Self-Test (bina dekhe likho, har question 5 marks)

- PAIN ke 4 aspects likho aur har ek ka ek example do.
- RAID 0, RAID 1 aur RAID 5 ka diagram banao aur explain karo. Comparison table do.
- Authentication ke teen factors kya hain? MFA kya hai?
- Authentication aur Authorization me difference likho.
- Anti-virus me signature-based detection kaise kaam karta hai? Uski limitation kya hai?

Self-Test Answers (likhne ke baad check karo)

1. Privacy (WhatsApp end-to-end encryption), Authentication (ATM PIN), Integrity (file download ke baad hash match), Non-repudiation (online payment par digital signature).

2. RAID 0: striping, min 2 disks, fault tolerance nahi. RAID 1: mirroring, min 2 disks, 1 failure bardasht, 50% capacity. RAID 5: striping + distributed parity, min 3 disks, 1 failure bardasht. Figure 6 aur Q2 ki comparison table banao.

3. Something you know (password, PIN), something you have (OTP, smart card), something you are (fingerprint, face). MFA ka matlab in me se 2 ya zyada ek saath use karna, jaise ATM card + PIN.

4. Authentication verify karta hai ki user **kaun** hai (login). Authorization decide karta hai ki login ke baad user **kya kar sakta hai** (permissions). Authentication pehle aata hai.

5. Ye har file ko known virus signatures (unique patterns) ke database se match karta hai. Match hone par file quarantine ya delete hoti hai. Limitation: signature database update hone tak naye (zero-day) viruses nahi pakad sakta.

> **Unit 1 complete!**
> Agar 5 me se 4 ya zyada sahi hue to **Unit 2** par jao. Nahi to Part A me jo topics miss hue unhe dobara padho.